Example Lab

> ⚠ **Synthetic example.** Built by running sigil on forage's mock collection. Every figure is a genuine output of that run, but the lab and its work are fabricated. It shows the format; it describes no real lab.

Overview for collaborators · Built 2026-11-04

> *We study how the heart and other organs repair themselves after injury, from the molecular programs that switch on in regenerating cells to the tissue architecture they rebuild. Most of our work pairs single-cell and spatial genomics with physiology and imaging in mouse injury models.*

Cardiac regeneration is the lab's largest theme, followed by disease modeling, tissue architecture and gene regulation. Most questions are answered with bulk or single-cell RNA-seq, alongside physiology, imaging, proteomics and behavior.

At a glance: 5 research themes · 40 sequencing projects · 15 project folders of data · 13 methods in use

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1. Research themes

THEME	PROJECTS	MAIN APPROACHES
Cardiac regeneration	13	Bulk RNA-seq (6), single-cell RNA-seq (5), DNA-seq (1), spatial (1)
Disease modeling	9	DNA-seq (3), bulk (2), single-cell (2), chromatin accessibility (1), multiome (1)
Tissue architecture	7	Single-cell RNA-seq (3), bulk (3), spatial (1)
Gene regulation	7	Bulk RNA-seq (4), single-cell RNA-seq (3)
Development	4	Bulk RNA-seq (2), single-cell (1), DNA-seq (1)

2. Methods beyond sequencing

Spatial transcriptomics, patch-clamp electrophysiology, calcium imaging, confocal microscopy, in situ hybridization, proteomics and behavior.

3. Biological focus

- **Tissues:** heart (left ventricle, cardiomyocytes), kidney, liver, lung, cortex.
- **Stages and timepoints:** mostly time after injury or treatment, with some neonatal and adult work.
- **Conditions and models:** myocardial infarction, injury, high-fat diet.

4. Toolkit

- **Genetic tools:** CreERT2 driver, tdTomato reporter, GCaMP indicator.
- **Analysis languages:** Python, R, MATLAB.

Built by sigil from the lab's data collection.